

KON 313

Feedback Control Systems

Introduction to Control Systems

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Introduction to Automatic Control

- Automatic control is essential in any field of engineering and science. It is desirable that most engineers and scientists are familiar with theory and practice of automatic control.
- Automatic control is an important and integral part of modern society. Numerous applications are all around us:

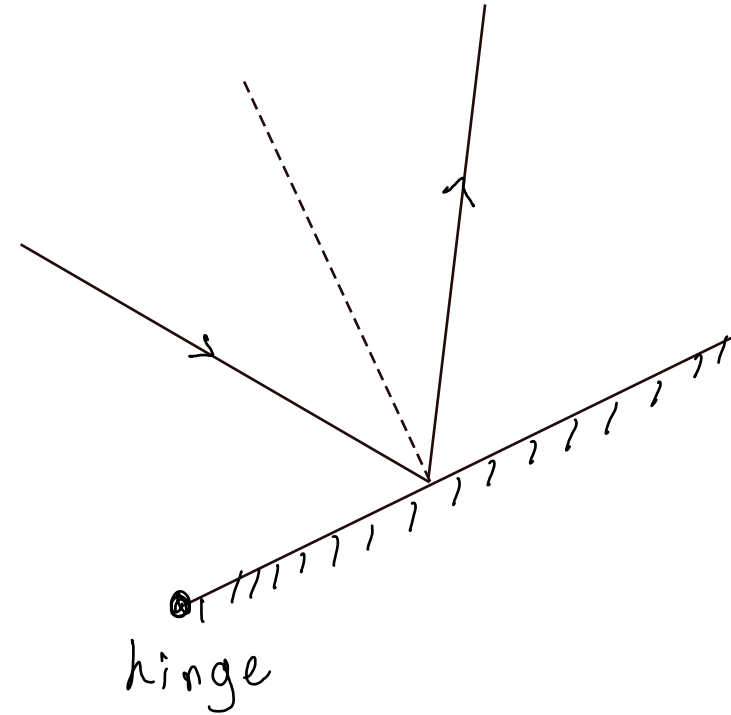
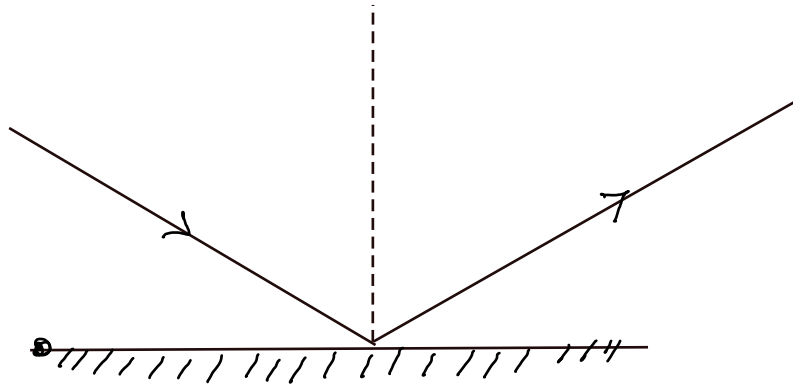
Control systems that we create: A self-guided vehicle delivering material by seeking its destination, Temperature/Humidity control systems

Natural Control Systems: Our eyes follow a moving object to keep it in view

Control systems in nonphysical world: Automatic control of a Student performance, to control the study time for a rise in grade

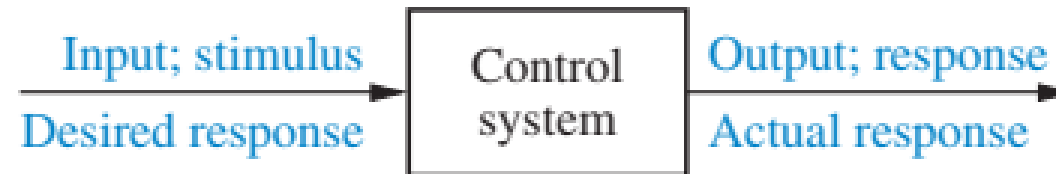
Introduction to Automatic Control

- A mirror that control the reflection of light direction is an example of an initial Control system.



Control System Definition

- A control system consists of *subsystems* and *processes* (or *plants*) assembled for the purpose of obtaining a desired *output* with desired *performance*, given a specified *input*.
- A control system in its simplest form:



Basic Concepts

- **System:** A combination of components that act together and perform a certain objective.
- **Controlled Variable (Output):** The quantity or condition that is measured and controlled. Normally, the controlled variable is the output of the system.
- **Input:** a signal supplied to the system that causes it to behave in a certain way. The reference input represents the desired value of the controlled output.
- **Plant (Controlled System):** A physical object, system or process that is to be controlled.

Basic Concepts

- **Control Signal:** The quantity or condition that is varied by the controller so as to affect the value of the controlled variable.
- **Control:** Measuring the value of the controlled variable of the system and applying the control signal to the system to correct or limit deviation of the measured value from a desired value.
- **Controller:** a device which adjusts the control signals according to a set of predetermined rules.
- **Disturbance:** A signal that tends to adversely affect the value of the output of a system.

Control Theories/Categories

- Linear vs Non-linear
- Time-invariant vs. Time-variant
- Open-loop vs. Closed-loop
- SISO (single input single output) vs. MIMO (multiple input multiple output)
- Classical (Conventional) vs. Modern
 - The **frequency-response** and **root-locus** methods are the core of classical control
 - Modern theory is based on **time-domain** analysis of differential equation systems
- Robust
 - Incorporates both the frequency-response and the time-domain approach
- Optimal, Adaptive, Stochastic (vs. deterministic), Digital, Intelligent Control

Linear Time-Invariant (LTI) Systems

- **Time-Invariant:**

Conceptually: The behavior and characteristics of the system are fixed over time.

In system language: A time shift in the input signal results in an identical time shift in the output signal.

$$\begin{cases} \text{if} & x(t) \longrightarrow y(t) \\ \text{then} & x(t-t_0) \longrightarrow y(t-t_0) \end{cases}$$

Ex. TI $\checkmark$ $y = \sin(x(t))$

$\times$ $y = tx(t)$, $y = x(2t)$

Linear Time-Invariant (LTI) Systems

- **Linear:**

The system possesses the *superposition* property.

$$x_1(t) \rightarrow y_1(t), \quad x_2 \rightarrow y_2(t)$$

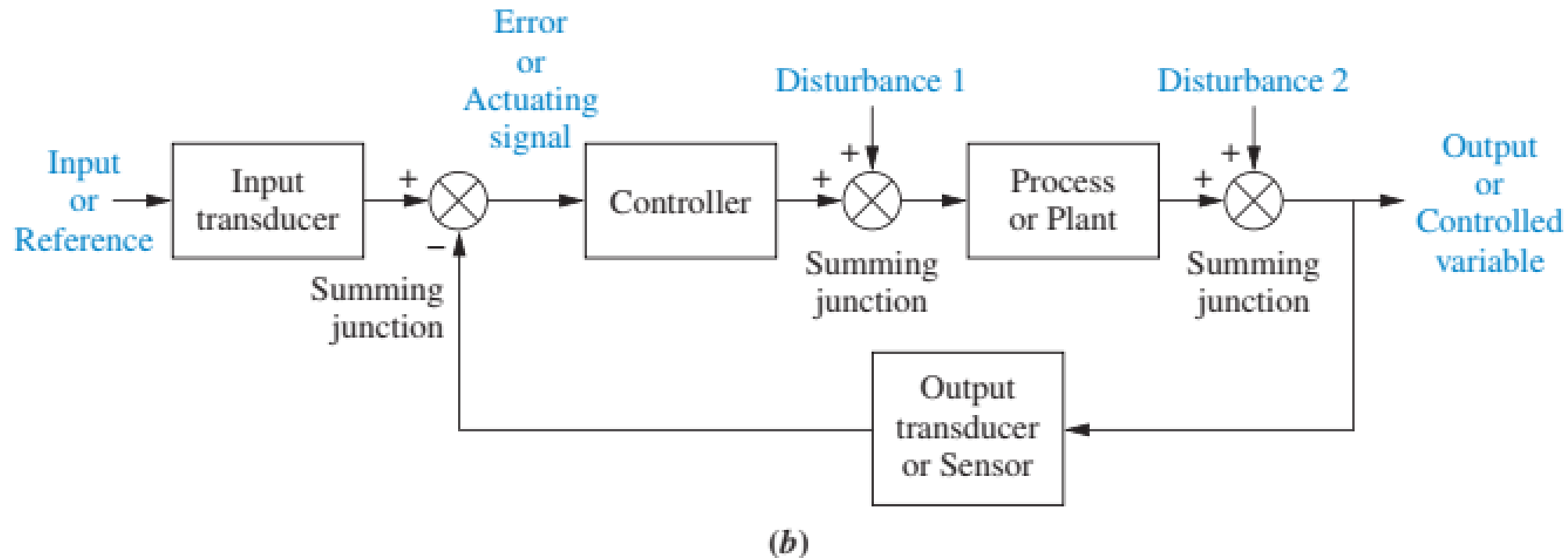
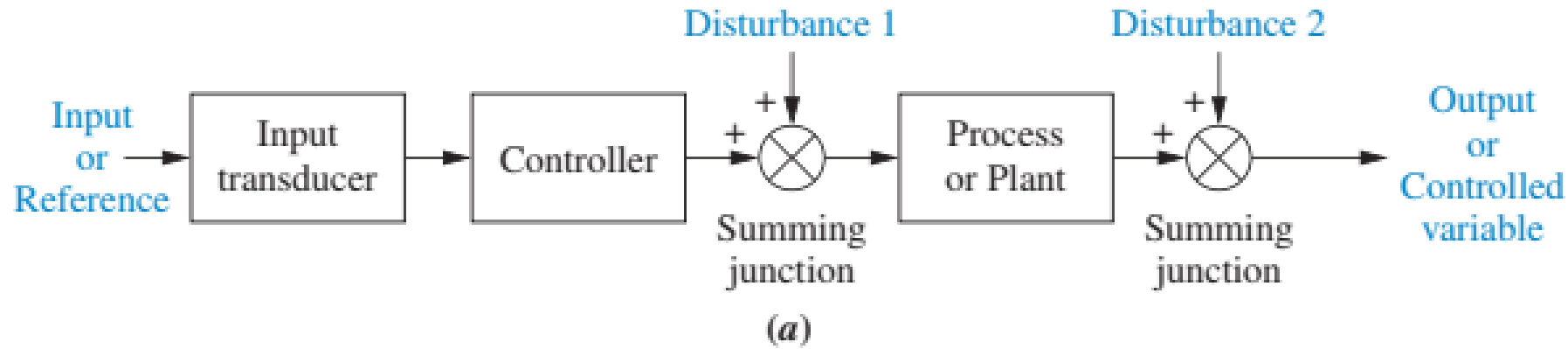
$$\Rightarrow x_1(t) + x_2(t) \longrightarrow y_1(t) + y_2(t)$$

$$\alpha x_1(t) \longrightarrow \alpha y_1(t), \quad \alpha \text{ is any complex constant}$$

Open Loop vs Closed Loop Systems

- **Closed-Loop (Feedback) Control:** Controls considering both input and output of the system. Tends to reduce the difference between the feedback signal (that may be the output of a system or a function of the output signal) and the reference input. E.g. cruise control found in cars, traffic control system, room temperature control, ...
- **Open-Loop Control:** The output has no effect on the control action. Open Loop Control Systems are not controlled by any other feedback or sensors but only work based on an input set point. E.g. washing machine, toaster, traffic light, ...

Open Loop vs Closed Loop Systems



Open Loop vs Closed Loop Systems

- **Open-Loop Control:**

- Simple,
- Low cost,
- Stability is not a major problem,
- Cannot function properly in the presence of disturbances and parameter variations.

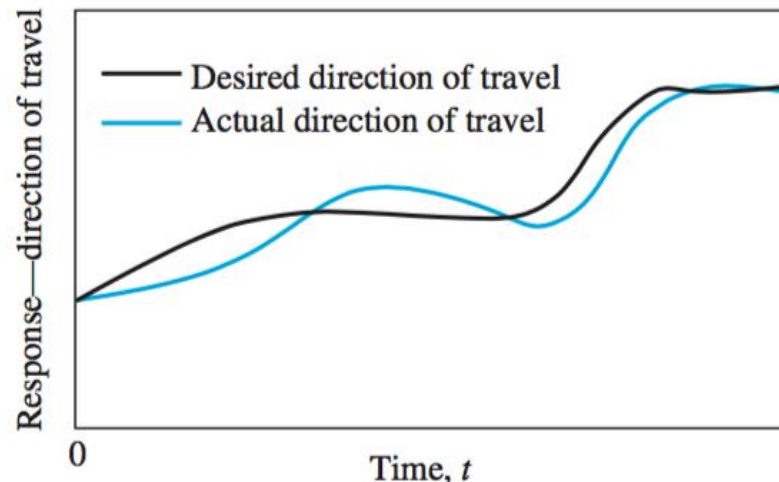
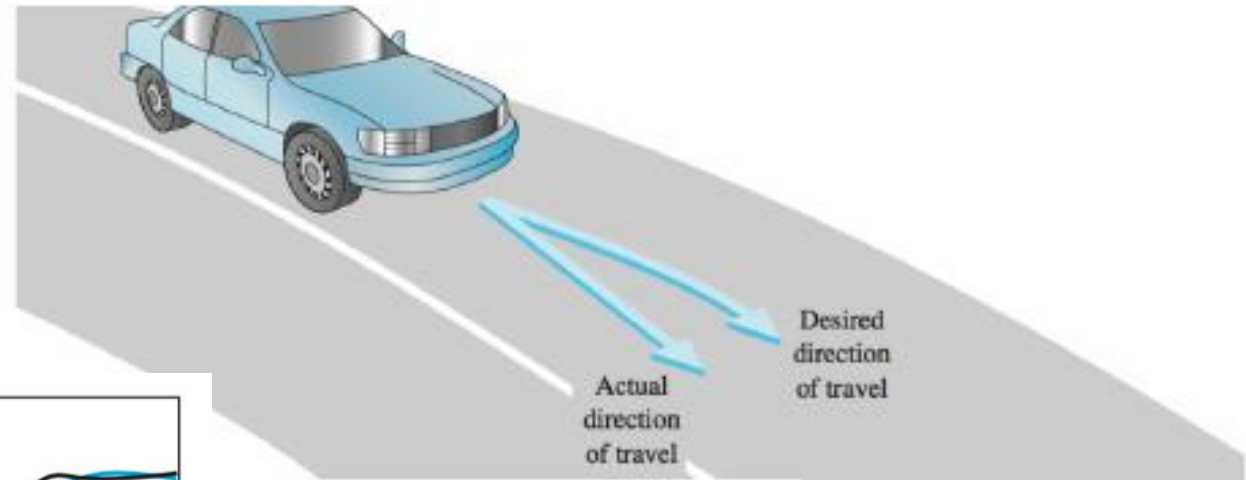
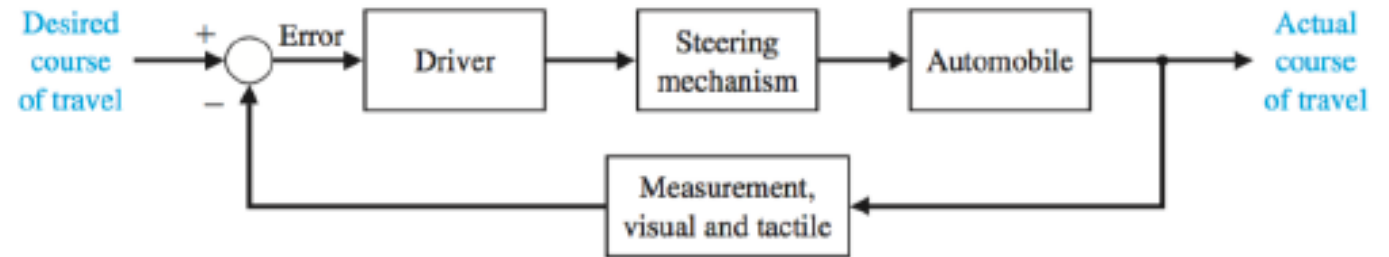
- **Closed-Loop (Feedback) Control:**

- More elements,
- Higher cost,
- Stability is a major problem,
- Less sensitive to disturbances and variations of parameter values.

An Examples of a Closed-Loop Control Systems

Automobile steering control system

- The driver uses the difference between the actual and the desired direction of travel to generate a controlled adjustment of the steering wheel.



Analysis, and Design of Control Systems

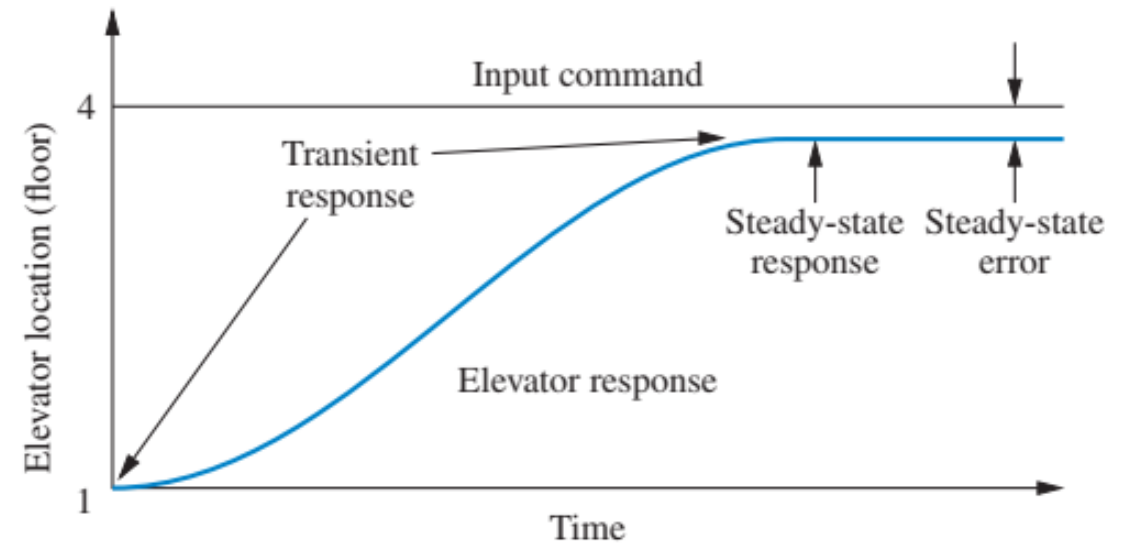
Analysis is the process by which a system's performance is determined.

Design is the process by which a system's performance is created or changed.

Analysis, and Design of Control Systems

Ex. (elevator):

- When the fourth-floor button is pressed on the first floor, the elevator rises to the fourth floor with a **speed** and **floor leveling accuracy** designed for passenger comfort.
- The push of the fourth-floor button is an **input** that represents our **desired output**.
- The level of the elevator from first floor is the **output**.
- The performance is determined by the elevator response containing:
 - (1) the **transient response** and
 - (2) the **steady-state response**.



Analysis, and Design of Control Systems

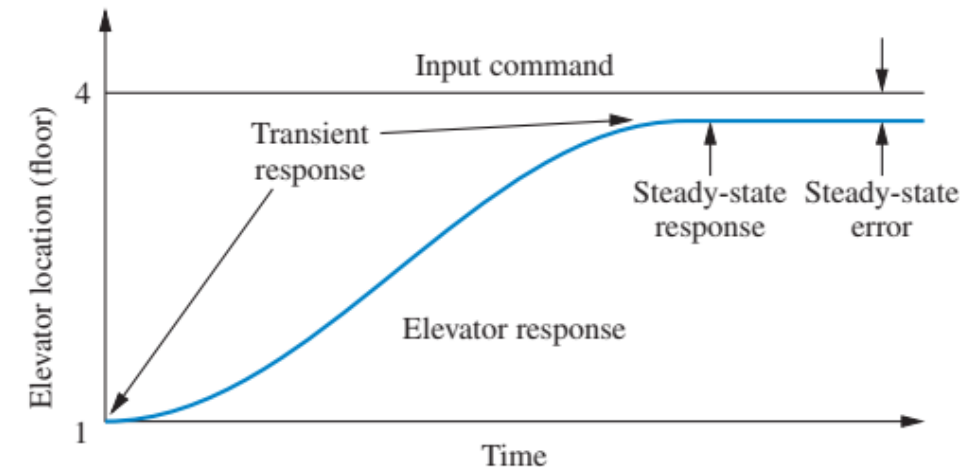
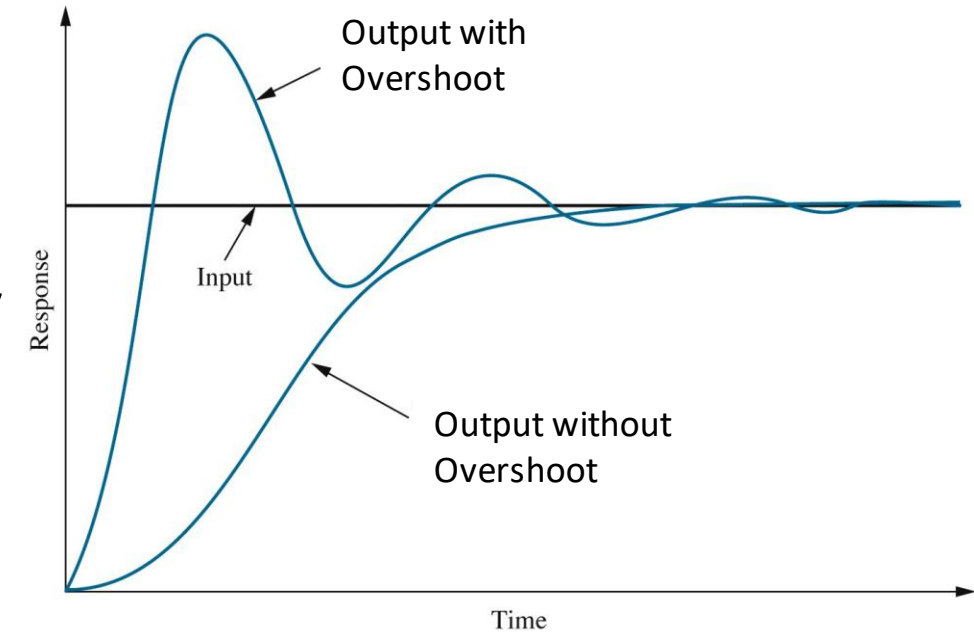
Analysis and Design Objectives

- Transient response
- Steady state response (transient response has decayed to zero remaining part resembles the input: output)
- Stability: the most important one in analysis and design objectives

Analysis, and Design of Control Systems

Analysis and Design Objectives

- Transient Response must meet certain criteria.
 - ✓ Mainly: Speed and maximum overshoot.
 - E.g. Elevator: a slow transient response makes passengers impatient, whereas an excessively rapid response makes them uncomfortable.
 - If the elevator oscillates about the arrival floor for more than a second, a disconcerting feeling can result.
- Steady-State Response must meet certain criteria.
 - ✓ Mainly: small or zero error
 - E.g. An elevator must be level enough with the floor for the passengers to exit.
- Stability must be guaranteed.



Analysis, and Design of Control Systems

Analysis and Design Objectives

- Stability must be guaranteed.

Total Response = Natural Response + Forced Response

Natural response describes the way the system dissipates or gain energy (*homogeneous* solution in linear differential equations). It is dependent only on the system not the input

Forced response depends on the input (*particular* solution in linear differential equations).

The natural response must (1) eventually approach zero, thus leaving only the forced response, or (2) oscillate.

If the natural response grows without bound, the system is no longer controlled (*instability*). E.g., the elevator would crash through the floor or exit through the ceiling;

Analysis, and Design of Control Systems

Analysis and Design Objectives

- Generally, the performance specifications should not be more stringent than necessary to perform the given task.
- For example, If the accuracy at steady-state operation is of prime importance in a given control system, then we should not require unnecessarily rigid performance specifications on the transient response, since such specifications will require expensive components.
- Remember that the most important part of control system design is to state the performance specifications precisely so that they will yield an optimal control system for the given purpose.

Analysis, and Design of Control Systems

The Design process → Developing a Mathematical Model (Block Diagram)

- The system should be modeled mathematically.
- The designer uses physical laws here, e.g.
 - Kirchhoff's laws for electrical networks, and
 - Newton's law for mechanical systems
- One such model is the linear, time-invariant *differential equation*,

$$\frac{d^n c(t)}{dt^n} + d_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \dots + d_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \dots + b_0 r(t)$$

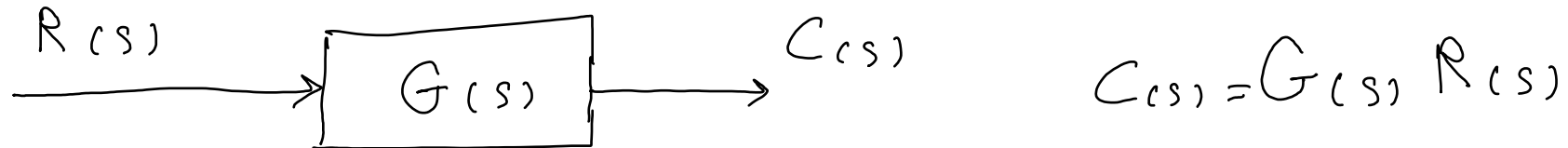
Analysis, and Design of Control Systems

The Design process → Developing a Mathematical Model (Block Diagram)

- *Transfer function* is another way of mathematically modeling a system.

The model is derived from the linear, time-invariant differential equation using the *Laplace transform*.

$$G(s) = \frac{C(s)}{R(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}$$

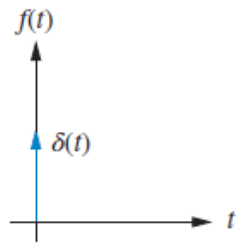
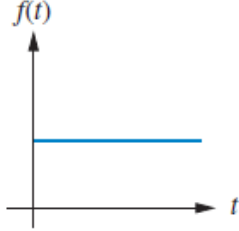
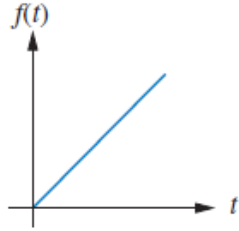
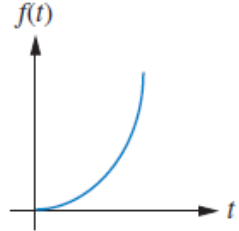
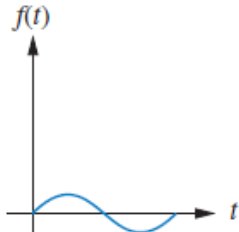


- Still another model is the *state-space representation*.

Basically, this representation turns an n th-order differential equation into n simultaneous first-order differential equations.

Analysis, and Design of Control Systems

- Test input signals are used to verify the design:

Input	Function	Description	Sketch	Use
Impulse	$\delta(t)$	$\delta(t) = \infty$ for $0- < t < 0+$ $= 0$ elsewhere $\int_{0-}^{0+} \delta(t) dt = 1$		Transient response Modeling
Step	$u(t)$	$u(t) = 1$ for $t > 0$ $= 0$ for $t < 0$		Transient response Steady-state error
Ramp	$tu(t)$	$tu(t) = t$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Parabola	$\frac{1}{2}t^2u(t)$	$\frac{1}{2}t^2u(t) = \frac{1}{2}t^2$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Sinusoid	$\sin \omega t$			Transient response Modeling Steady-state error

Analysis, and Design of Control Systems

The approaches that we will use to control system design

- ✓ Frequency-response approach,
- ✓ The root-locus approach,
- ✓ The state-space approach,
- ✓ The PID-based compensational approach

Q1-1

Which option most correctly describes a closed loop system?

- a) Conventional toaster
- b) Car cruise control
- c) Room heating system
- d) Conventional traffic light

Answer: b